

Relax with Apache CouchDB



Traditional relational databases are not designed to cope with the varying agility and scaling requirements of modern day applications; nor can they be made to run on cheap, readily available commodity hardware. This has raised the need to not only change the way databases work, but also how they scale and function. Apache CouchDB addresses these needs.

A NoSQL or a 'Not only SQL' database is a non-relational, largely distributed database system that can store and analyse very complicated and unstructured data with ease. It addresses the issues that are faced by modern day applications when using traditional databases.

NoSQL database types

NoSQL comprises a few different types of database technologies, each with its own design, function and set of uses.

- **Document databases:** This category of NoSQL databases pairs each key with a complex data structure called a document, which can store and process data in many different arrangements based on the requirements. These arrangements include key-value pairs, key-array pairs or even a complex data structure such as a nested document.
- **Key-value databases:** These are the most simple and widely used NoSQL database technologies. Each object or item in the database is stored with a unique attribute name, also called as key, together with its value. Key-value databases combine a few elements from traditional

relational databases as well, such as the ability to provide a value with a type to perform more data operations over it, like an integer or a float.

- **Wide-column databases:** These are designed to cope with tremendous volumes of data that can span across millions of columns. These databases, notably, Cassandra and Hadoop HBase, are optimised for processing queries over large datasets, where data is grouped and stored in columns rather than rows.
- **Graph databases:** These are based on the concept of Graph Theory. They are designed for data that is well represented and interconnected. Graph databases are commonly used to store information obtained from social media connections, real-world applications, etc.

Table 1 briefly compares some of the most common and frequently used NoSQL databases.

The benefits of NoSQL

Now that we have understood what a NoSQL database is, let us have a quick look at some of its key features and benefits compared to traditional relational databases.